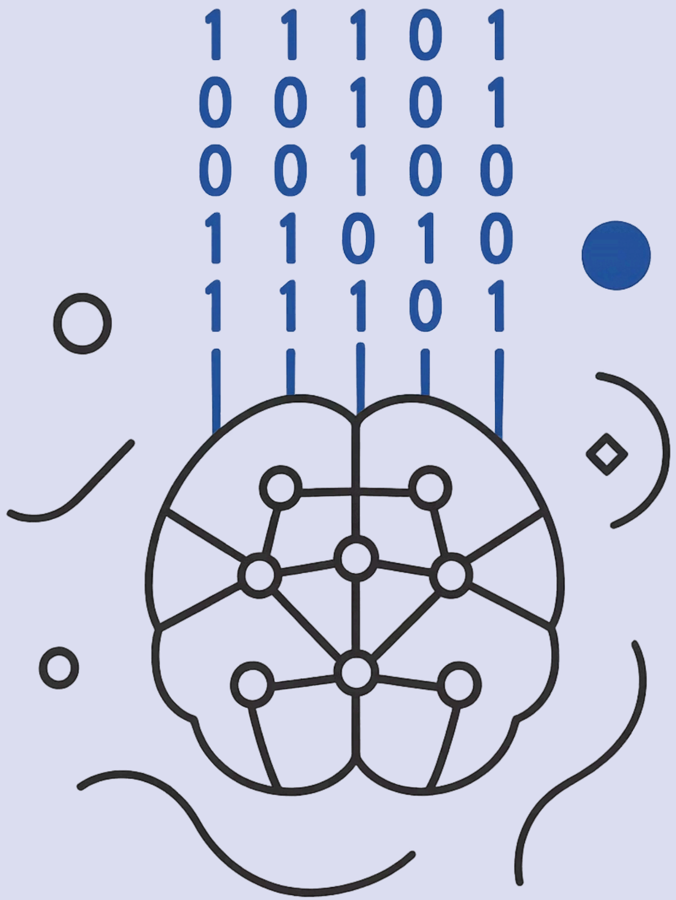
Entering the AI World

From Data to Intelligence

SECTION 1: AI WORLD AND DATA SCIENCE



What We Will Cover

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Why AI Became Powerful Now

02

The AI Family Tree

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Where Data Science Fits

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The Data Science Lifecycle

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Transition to Hands-On Practice

Did AI Start With ChatGPT?

Many people think AI is a recent invention.

But the dream of making machines intelligent has existed for many decades.

> AI did not start with ChatGPT.
>
> What changed recently is that the <u>conditions</u> became right.

The AI Dream Was Old

AI has been researched and attempted for decades.

Earlier AI was held back by fundamental limitations.

1. **Less Data**

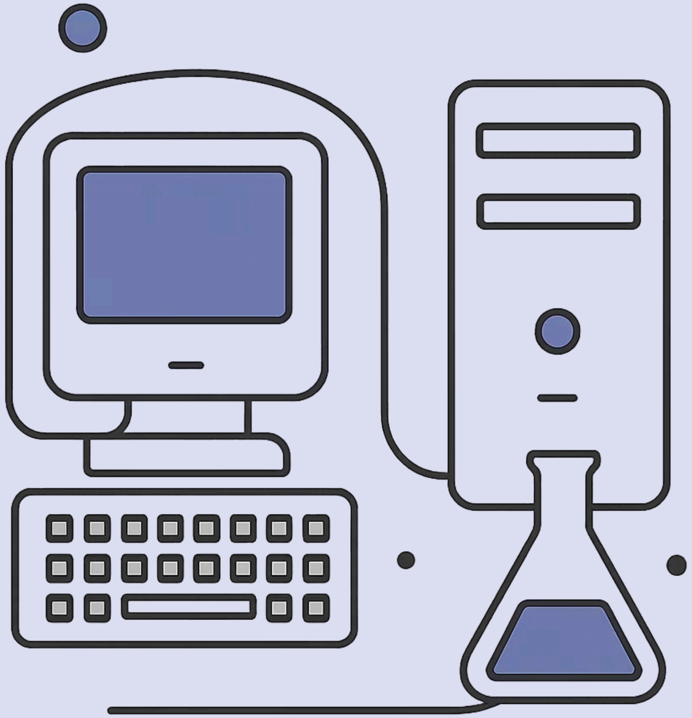
   Digital activity was limited

2. **Immature Algorithms**

   Learning methods were not ready

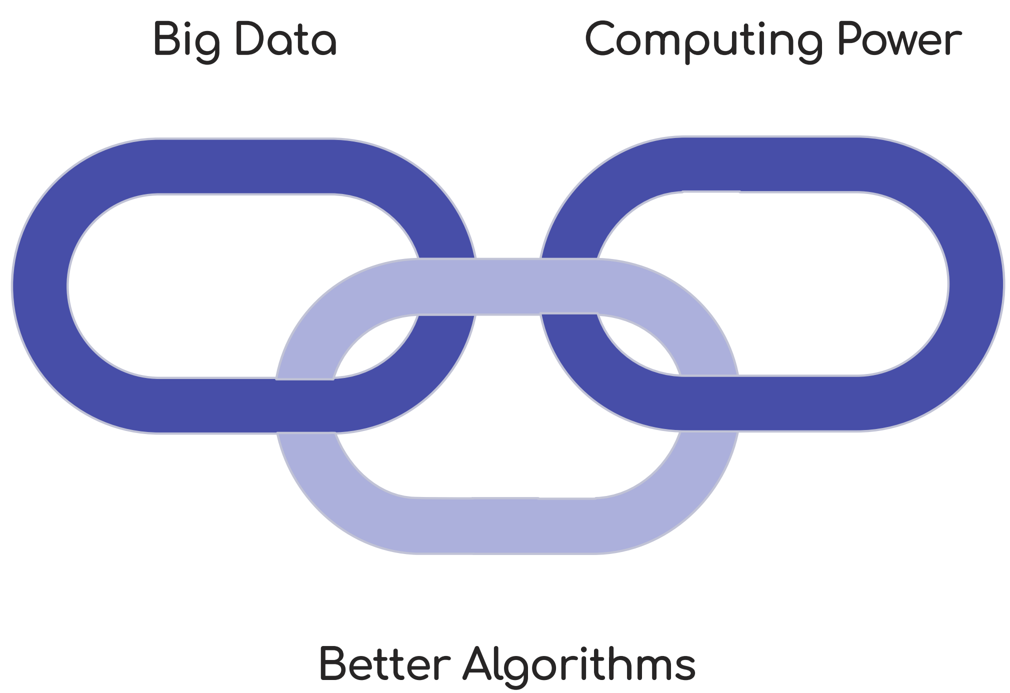
3. **Weak Hardware**

   Computing was slow and expensive



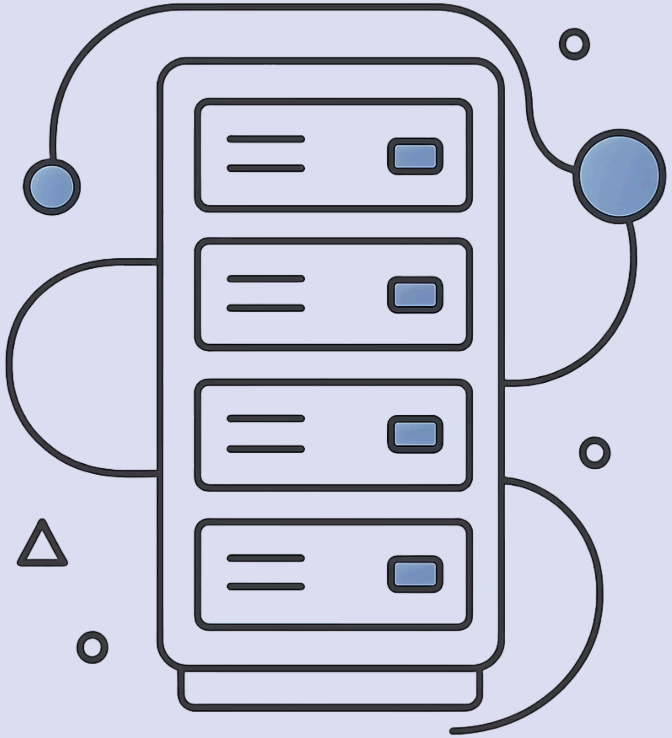
The Timing Became Right

Modern AI became powerful because <u>three forces</u> matured together.



AI did not become powerful because of one invention.

It became powerful because the full ecosystem became ready.



Force 1: Big Data

Data is the fuel.

Everyday digital activity generates massive examples AI can learn from.

Sources

Mobile apps, social media, payments, GPS, sensors, hospitals, video platforms, ...

Why It Matters

More examples reveal better patterns.

Poor data creates poor AI behavior.

Without data, AI has nothing to learn from.

Force 2: Algorithms

Algorithms are the brain.

An algorithm is a step-by-step method used to solve a problem.

In AI, algorithms help machines learn patterns from data.

- Machine Learning
- Deep Learning

Force 3: Computing Power

Computing power is the muscle.

Earlier, large AI experiments were slow and expensive.

Now we have the infrastructure to scale.

GPUs & TPUs

Specialized AI hardware

Cloud Computing

Scalable on-demand resources

Distributed Training

Parallel model training at scale

Open-Source Tools

Accessible AI frameworks

Why AI Feels Sudden

The public experienced AI directly through chatbots, image generation, and code assistants

> but the foundation was built over time.

AI Is Already Around Us

Which of these did you use in the last 24 hours?

Recommendations

Products, videos, courses

Voice Assistants

Siri, Alexa, Google

Fraud Detection

Banking and payments

Route Suggestions

Maps and navigation

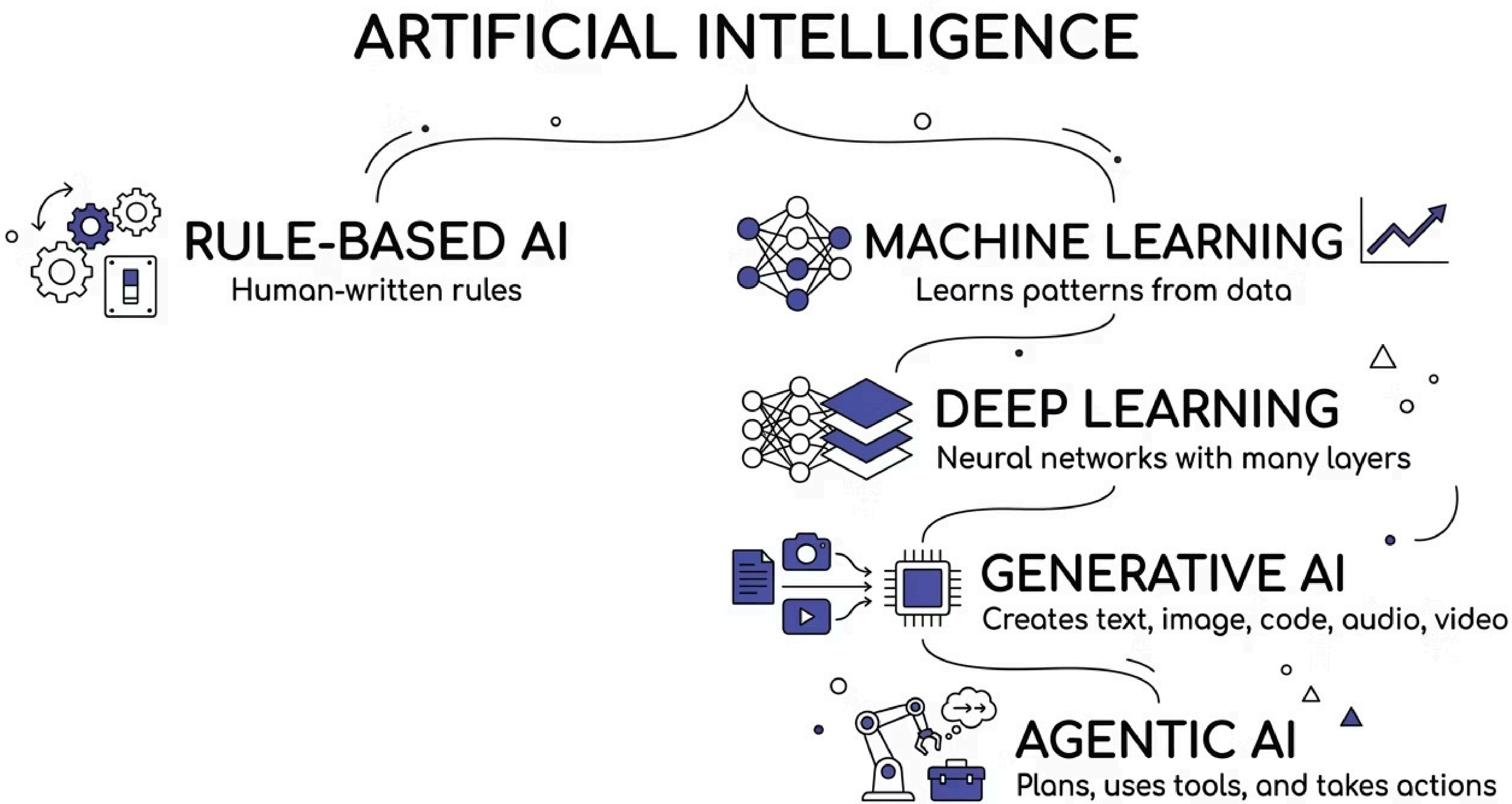
Chatbots

Support and assistants

Search Ranking

Web and app search

The AI Family Tree



This is a teaching map — not a strict taxonomy. It helps place concepts correctly before going deeper.

Rule-Based AI

How It Works

Humans write every rule explicitly.

The machine follows instructions — it does not learn.

```
if marks >= 35:
    result = "Pass"
else:
    result = "Fail"
```

Good For

- Fixed decisions
- Clear business rules
- Compliance checks

Weak For

- Complex patterns
- Image, speech, language
- Ambiguous or changing cases

Machine Learning

Instead of writing every rule, we give examples.

The algorithm learns which patterns connect to outcomes.

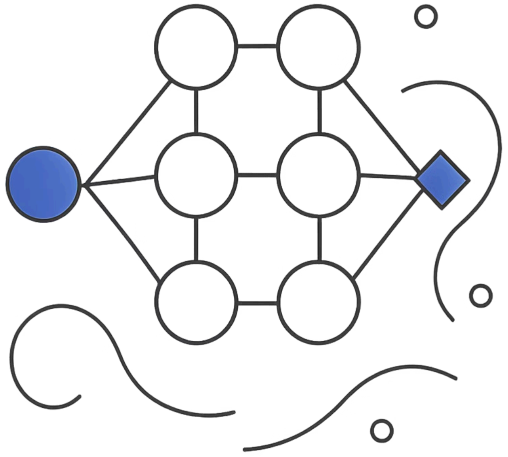
Example:

Past student data — attendance + assignment score + lab score

→ the model learns patterns connected to the final outcome.

Good for prediction, classification, pattern recognition, risk scoring, and recommendation.

Deep Learning



Neural Networks With Many Layers

Deep Learning handles complex data types that simpler models struggle with.

- Images and video
- Audio and speech
- Natural language
- Medical scans

Deep Learning became powerful when large data and strong computing power became available.

Generative AI

Generative AI creates new content — not just predictions.

Traditional ML	Generative AI
Predicts a value or category	Creates new content
Example: predict student marks	*Example: write an explanation, generate an image, produce code*

Traditional AI predicts. Generative AI creates.

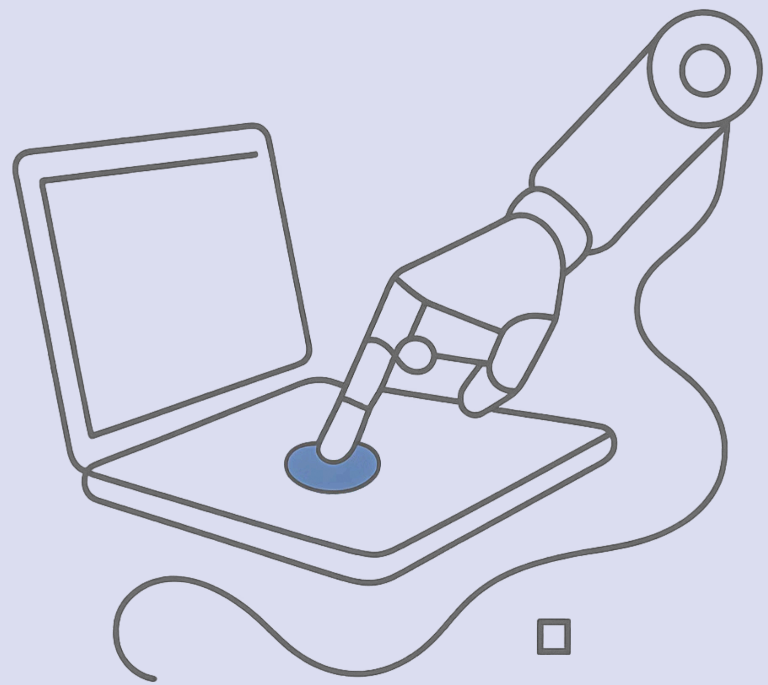
Agentic AI

Agentic AI moves from answering to acting.

1. Understand Goal
2. Break Into Steps
3. Use Tools & APIs
4. Act & Verify

Example: A normal chatbot tells you how to book a ticket.

An AI agent searches, compares, confirms, books, and sends a summary.



Data Science vs. AI

Data Science Focuses On

- Understanding and cleaning data
- Exploring patterns
- Statistics and visualization
- Communicating insights
- Supporting decisions

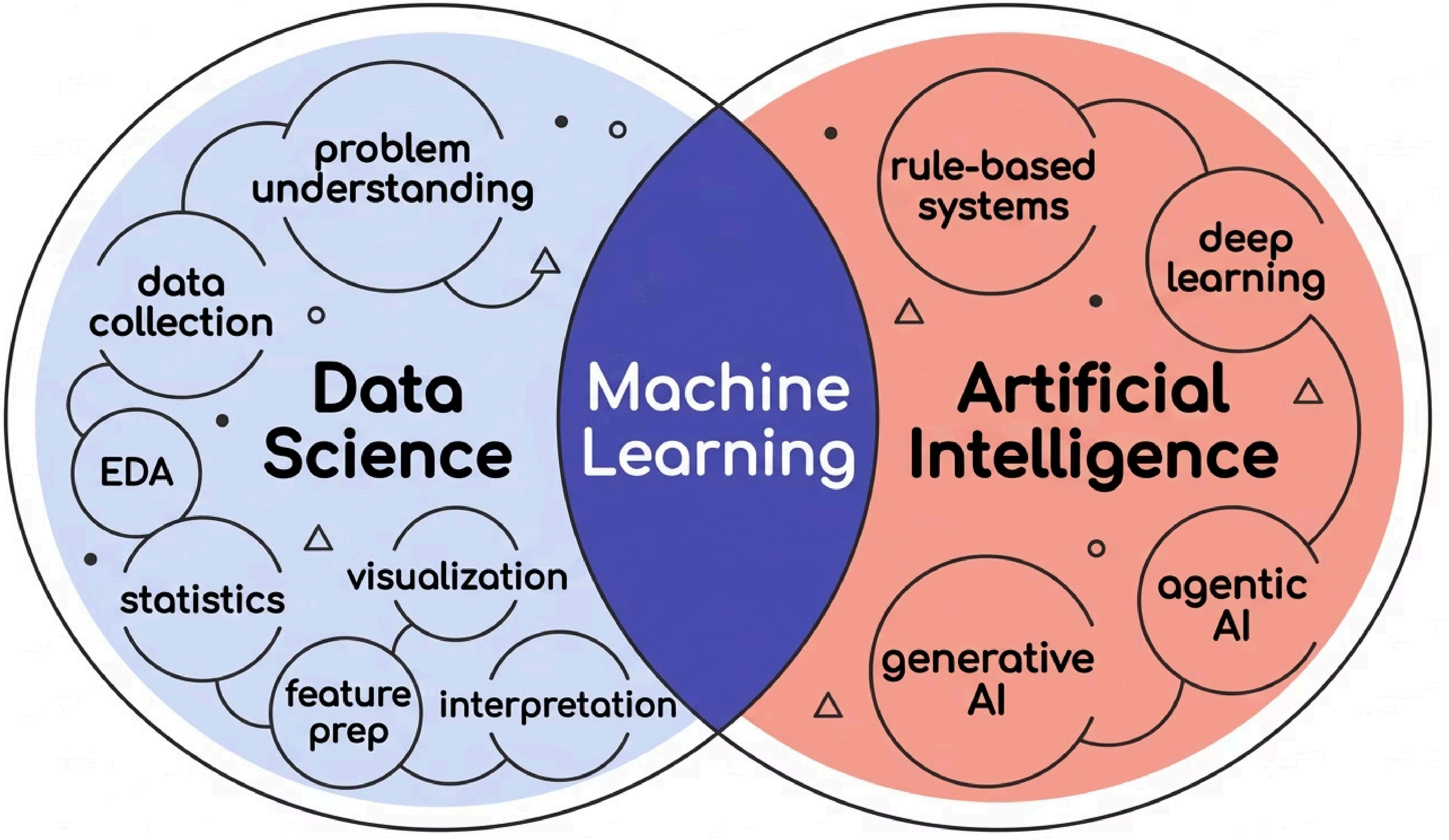
AI Focuses On

- Making machines behave intelligently
- Predicting and understanding
- Generating content
- Planning and acting

Data Science prepares and extracts value from data.

AI uses data and algorithms to create intelligent behavior.

The Relationship Map



Machine Learning is the major overlap between Data Science and AI — it lives in both worlds.

What Is Data Science?

> Data Science uses data to answer questions, find patterns, support decisions, and build data-driven solutions.

It combines domain knowledge, data cleaning, analysis, and clear communication.

Data Science turns raw data into useful decisions.

Model Building Is Only One Part

Many beginners think Data Science means only machine learning models.

Real Data Science is broader.

Business Understanding

Define the real problem before touching data

Data Quality

Exploratory analysis and cleaning

Storytelling

Communicating findings clearly

Responsible Decisions

Ethics, fairness, and monitoring

⚠ In real projects, understanding and preparing data often takes more effort than training the model.